

# INDUWARA DINUSITH WIJESINGHE

Undergraduate Mechanical Engineering Student — Mechatronics Specialization

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Portfolio - <https://induwarawijesinghe369-bit.github.io/> | GitHub - <https://github.com/induwarawijesinghe369-bit>

## Profile

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Third-year Mechanical Engineering undergraduate at the University of Moratuwa (2023 intake), specializing in the Mechatronics stream, with hands-on experience in robotics, mechanical design, simulation, embedded systems, and automation. Experienced in multidisciplinary engineering projects using SolidWorks, AutoCAD, MATLAB/Simulink, Simscape Multibody, Python, C++, Raspberry Pi, Arduino/ESP32, and computer vision. Seeking an engineering internship to apply knowledge of mechanical engineering, mechatronics, robotics, control, and simulation to practical engineering challenges.

## Education

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### BSc (Hons) in Mechanical Engineering (Mechatronics Specialization)

2024–Present

University of Moratuwa

GPA: 3.82 / 4.00

Relevant coursework: Mechatronics, Robotics, Automation, Mechanical Design, Engineering Mechanics, Dynamics and Control, Manufacturing.

### Nalanda College, Colombo 10

Advanced Level (2022): Z-score 2.1310 — Combined Mathematics A, Physics A, Chemistry A, General English A.

Ordinary Level (2019): 9 A's (Distinction passes).

## Technical Skills

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**Mechanical / Mechatronics:** Mechanical design, CAD modelling, kinematics, kinetics, fluid machinery fundamentals, robotics, system modelling, engineering problem solving

**Simulation / Engineering:** SolidWorks, AutoCAD, MATLAB, Simscape Multibody, COMSOL Multiphysics, MSC Adams, Ansys

**Programming / Computer Vision:** Python, C++, MATLAB programming, OpenCV, YOLO, basic image processing and computer vision

**Embedded / Hardware:** Raspberry Pi, ESP32, Arduino, sensors

**Transferable:** Teamwork, research, presentation, time management, problem solving

**Languages:** English (fluent), Sinhala (native)

## Engineering Projects

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### SAUVC 2026 — Autonomous Underwater Vehicle

IMU, Depth Control, ESP32/Firebase

- Contributed to development of an AUV for the SAUVC 2026 competition, focusing on underwater sensing, orientation, depth control, and communication.
- Integrated MPU9250/MPU6500 IMU and QMC5883L magnetometer for pitch, roll, yaw, and heading; used a pressure transmitter for depth control.
- Developed the ESP32/Firebase communication interface for transmitting control data to the onboard controller.

### Vision-Guided Cartesian Pick-and-Place Robot

Raspberry Pi, YOLOv8, OpenCV, Arduino

- Designed and built a Cartesian pick-and-place robot with camera-based object detection and positioning.
- Developed and trained a YOLOv8 model to identify target objects and obtain their coordinates using Raspberry Pi and Python/OpenCV.
- Implemented NEMA 17 stepper motor control using Arduino and A4988 drivers for the turntable and robot axes.
- Built a web-based control interface using React, JavaScript, HTML, and CSS; designed critical mechanical parts in AutoCAD.

### Smart Bicycle Instrumentation System

ESP32, IMU, GPS, Firebase IoT

- Designed an instrumentation system integrating heart-rate, IMU, GPS, and custom-built velocity sensors for real-time bicycle and rider data.
- Calibrated and filtered heart-rate sensor measurements using C++ to improve accuracy and reliability.
- Implemented GPS-based location tracking and ESP32 sensor data acquisition.
- Developed an IoT data pipeline and web GUI for real-time display of sensor data through Firebase.

## Reverse Engineering of an Infrared Cooker

Structural & Material Analysis

- Conducted reverse engineering of a metallic cover assembly, including material identification, manufacturing-process evaluation, and design improvement.
- Identified low-carbon steel (AISI 1006–1010) using spark testing, density measurement, and Optical Emission Spectrometry.
- Modelled the cover, punch, and die in CAD and proposed reinforcement ribs and hemmed edges.

## Multi-Purpose Device Holder

Manufacturing Technology

- Designed and fabricated a compact holder for a laptop, tablet, phone, and headset using wood and metal components.
- Applied woodworking and metalworking processes including cutting, drilling, mortise-and-tenon joinery, bending, and welding.
- Built an adjustable frame incorporating hinges and a wing-nut mechanism.

## Biped Wall-Climbing Robot

Mechatronic System Design Project — Ongoing

- Designing a 5-DoF biped wall-climbing robot using suction-based adhesion for vertical wall inspection.
- Developing a negative-pressure adhesion system and two-legged inchworm gait for controlled vertical locomotion.
- Developing DH-based kinematic and inverse-kinematic models for joint control and suction-unit positioning.
- Integrating cameras, ToF sensors, encoders, IMU, and pressure sensors for perception, feedback, orientation, and adhesion monitoring.

## 6-DOF Robotic Arm Simulation

MATLAB/Simulink, Simscape Multibody — Ongoing

- Developing a simulation model of a 6-DOF robotic manipulator to study robot kinematics and multibody motion.

## Professional Memberships & Achievements

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- Student Affiliate Member — Institution of Mechanical Engineers (IMechE), 2025–Present (Membership No. 80794909).
- Dean's List — University of Moratuwa, Faculty of Engineering (Semesters 2, 3, and 4).
- National Black Belt (1st Dan), Karate-do — Sri Lanka Karate-do Federation, awarded 2018.
- 2nd Place — Shotokan Karate-do National Championship Tournament, 2016.

## Certifications

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- Assured Diploma in Information Technology (DITEC) — ESOF Metro Campus (Pearson Assured), 2020.
- MATLAB Onramp — MathWorks.
- Simulink Onramp — MathWorks.
- Diploma in Spoken English (Merit Pass) — British Way English Academy.

## References

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### Dr. L. U. Subasinghe

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